Customer Behaviour Analysis Report

1. Project Overview

This project focuses on analysing customer purchasing behaviour using a combination of Python, SQL, and Power BI. The objective is to derive actionable insights related to customer preferences, spending habits, and segmentation, which can help businesses improve decision-making and marketing strategies.

2. Dataset Description

The dataset consists of approximately 3,900 customer transactions with multiple attributes capturing customer demographics, purchase details, and behavioral patterns.

- Customer Demographics: Age, Gender, Location, Subscription Status
- Transaction Details: Product Category, Item Purchased, Purchase Amount
- Behavioral Data: Frequency of Purchases, Previous Purchases, Discounts Applied
- Additional Attributes: Shipping Type, Ratings, Season, Size, Color

Data Quality:

- Minor missing values were observed and handled during preprocessing
- Data was cleaned and standardized to ensure consistency

3. Exploratory Data Analysis (Python)

Python was used extensively for data cleaning, preprocessing, and initial exploration.

Steps Performed:

- Data Loading: Imported dataset using pandas
- Initial Exploration: Used .info() and .describe() to understand structure and statistics

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Pay M
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	V
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	I
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	I

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

- Data Cleaning:
  - Handled missing values using suitable imputation techniques
  - Ensured consistency in column naming and formatting
- Feature Engineering:
  - Created new variables such as age groups
  - Derived purchase frequency indicators
- Trend Analysis:
  - Identified patterns in spending behavior
  - Analyzed distribution of purchases across categories
- Preparation for SQL:
  - Cleaned dataset was exported and stored in a database for further querying

4. Data Analysis using SQL

SQL was used to perform deeper business-oriented analysis and extract meaningful insights.

1. **Revenue Analysis:** Compared revenue contribution across different customer groups

Result Grid | Filter R

gender	revenue
Male	157890
Female	75191

2. High-Spending Discount Users :

Identified customers who used discounts but still spent above the average purchase amount.

Result Grid | Filter Rows:

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68

3.Top 5 Products by Rating:

Found products with the highest average review ratings.

Result Grid | Filter Rows:

item_purchased	Average Product rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

4. Shipping Type Comparison:

Compared average purchase amounts between Standard and Express shipping.

Result Grid | Filter Rows: | Exp

shipping_type	ROUND(AVG(purchase_amount),2)
Express	60.48
Standard	58.46

5. Subscribers vs. Non-Subscribers:

Compared average spend and total revenue across subscription status

Result Grid | Filter Rows: | Export: | Wrap Cell C

subscription_status	total_revenue	avg_spend	total_customers
Yes	62645	59.49	1053
No	170436	59.87	2847

6. Discount-Dependent Products:

Identified 5 products with the highest percentage of discounted purchases

Result Grid | Filter Rows:

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation:

Classified customers into New, Returning, and Loyal segments based on purchase history

Result Grid | Filter Rows:

customer_segment	number of customers
Loyal	3116
Returning	701
New	83

8. Repeat Buyers & Subscriptions:

Checked whether customers with >5 purchases are more likely to subscribe

Result Grid | Filter Rows:

subscription_status	repeat_buyers
Yes	958
No	2518

9. Revenue by Age Group:

Calculated total revenue contribution of each age group

Result Grid | Filter Rows:

subscription_status	repeat_buyers
Yes	958
No	2518

5. Dashboard in PowerBI

6. Business Recommendations

1. Boost Subscriptions – Promote exclusive benefits for subscribers.
2. Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.

3. Review Discount Policy – Balance sales boosts with margin control.
4. Product Positioning – Highlight top-rated and best-selling products in campaigns.
5. Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping user